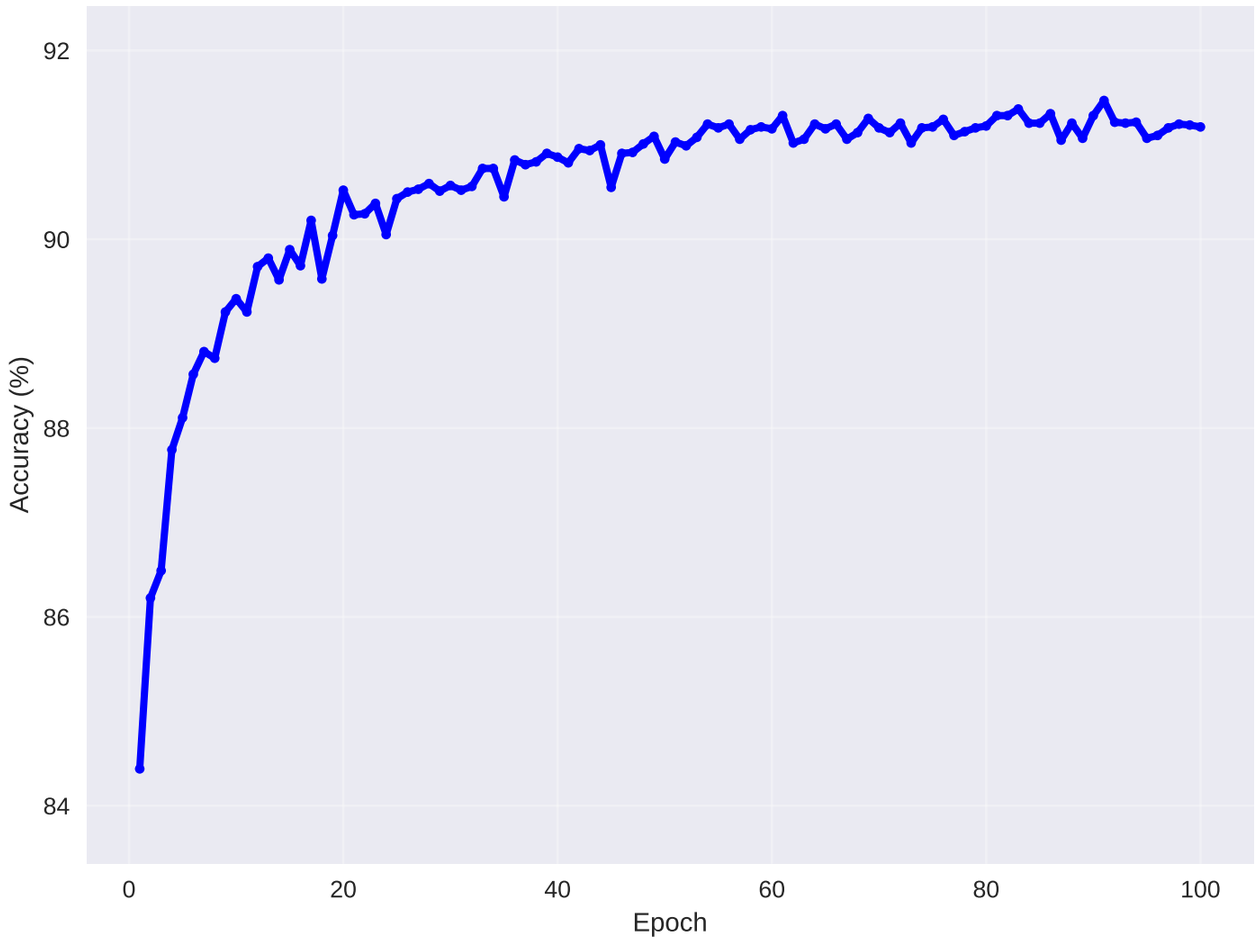
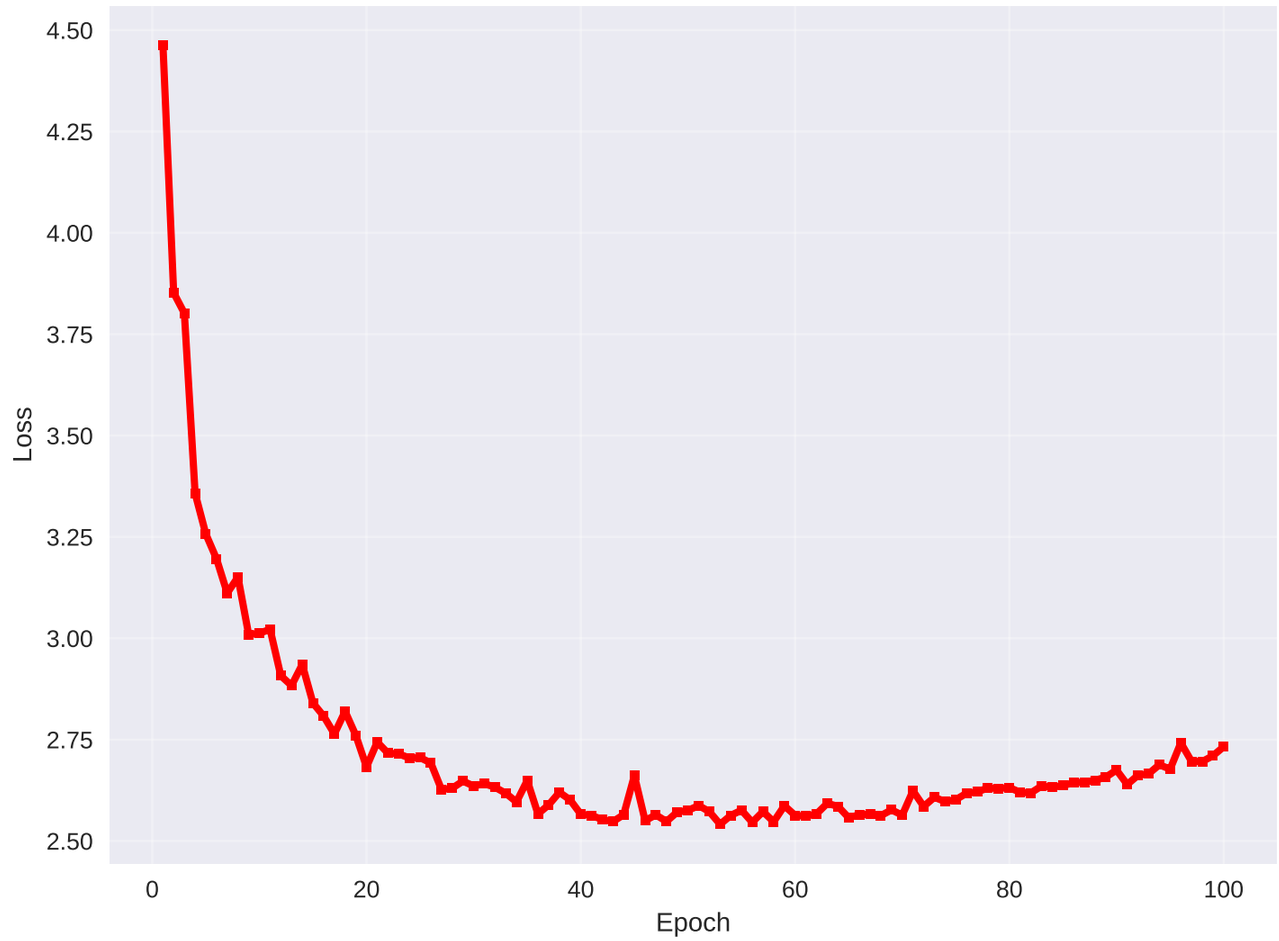
# Federated Learning - Main Metrics Overview

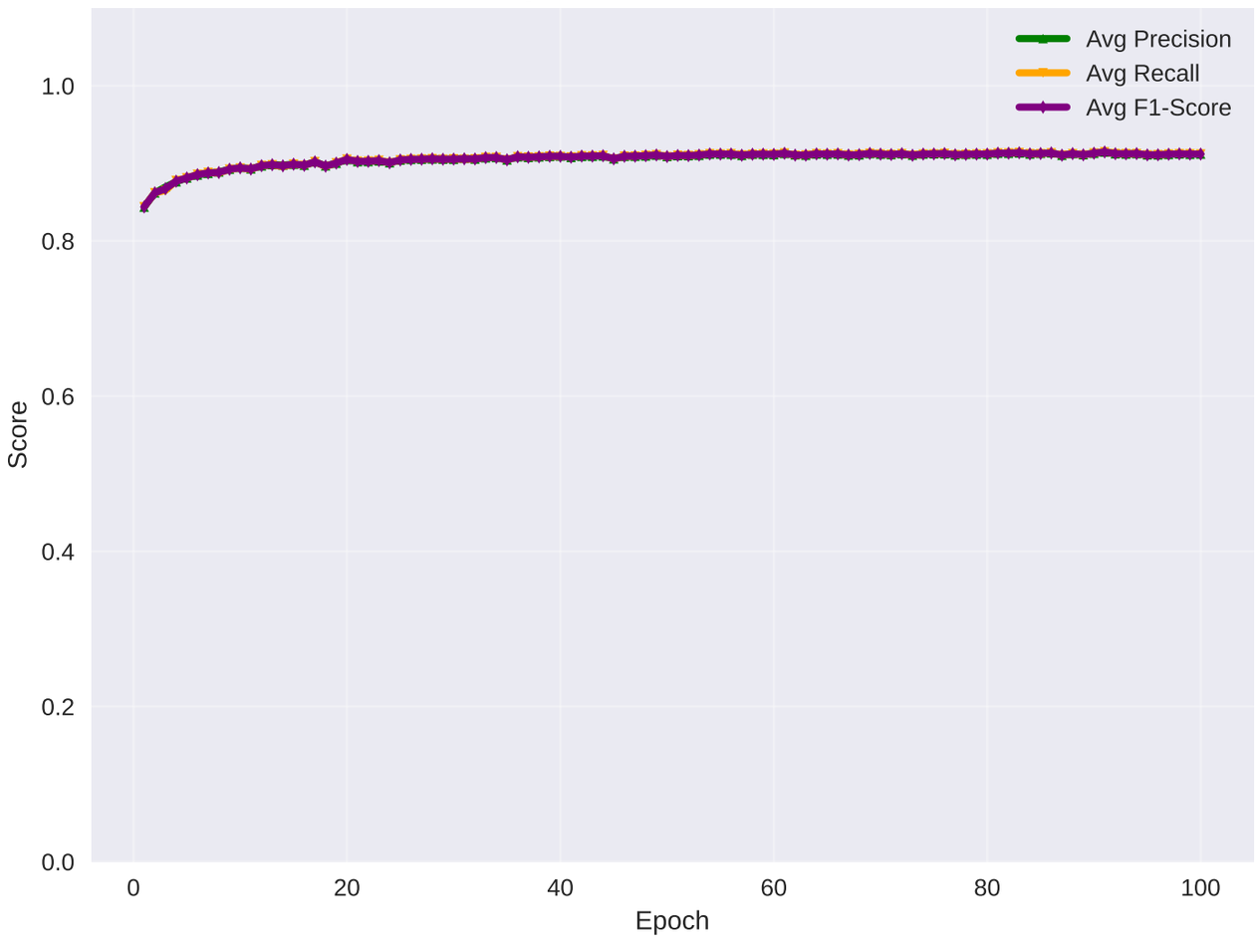
 Global Model Accuracy Evolution



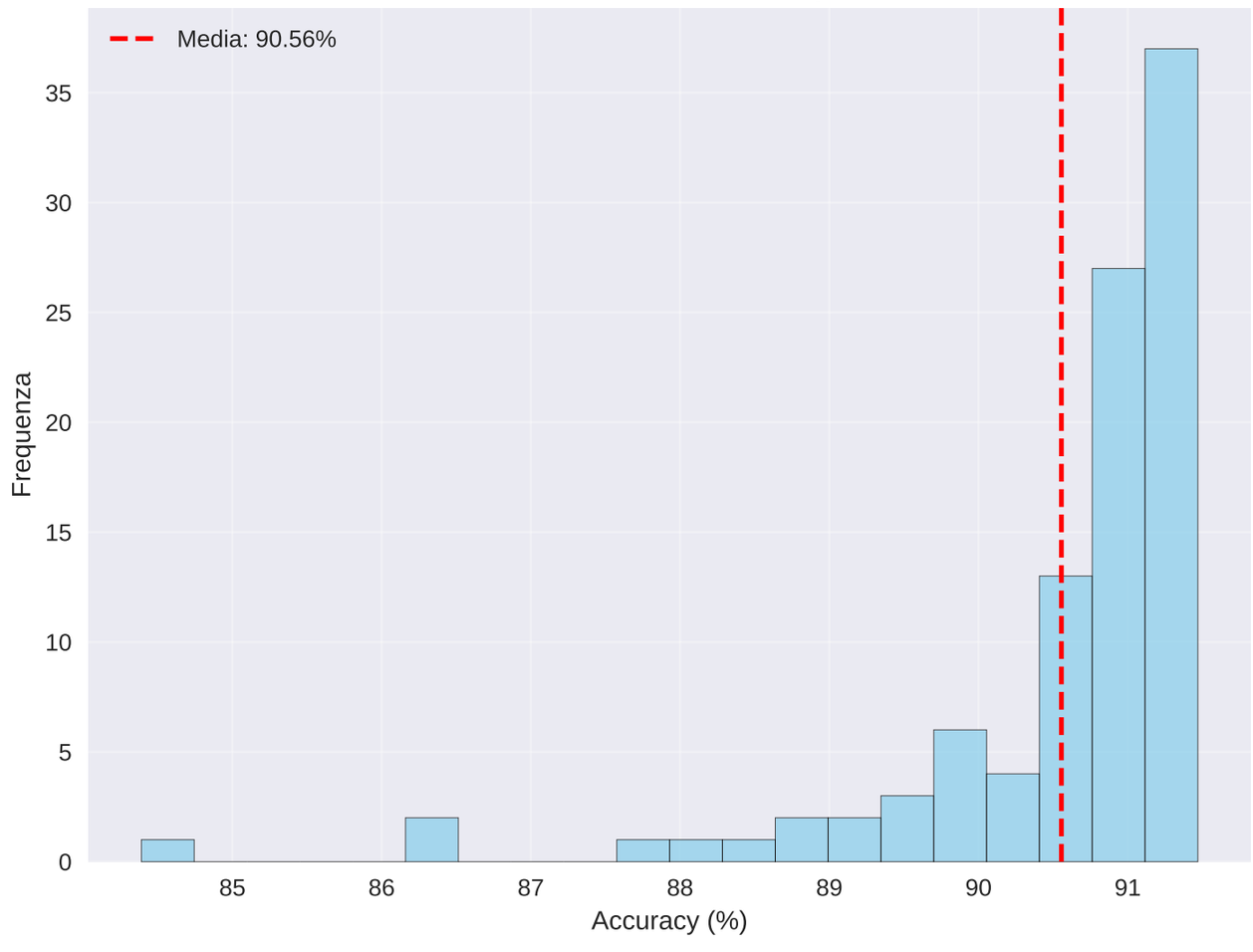

 Global Model Loss Evolution




 Average Precision, Recall & F1-Score

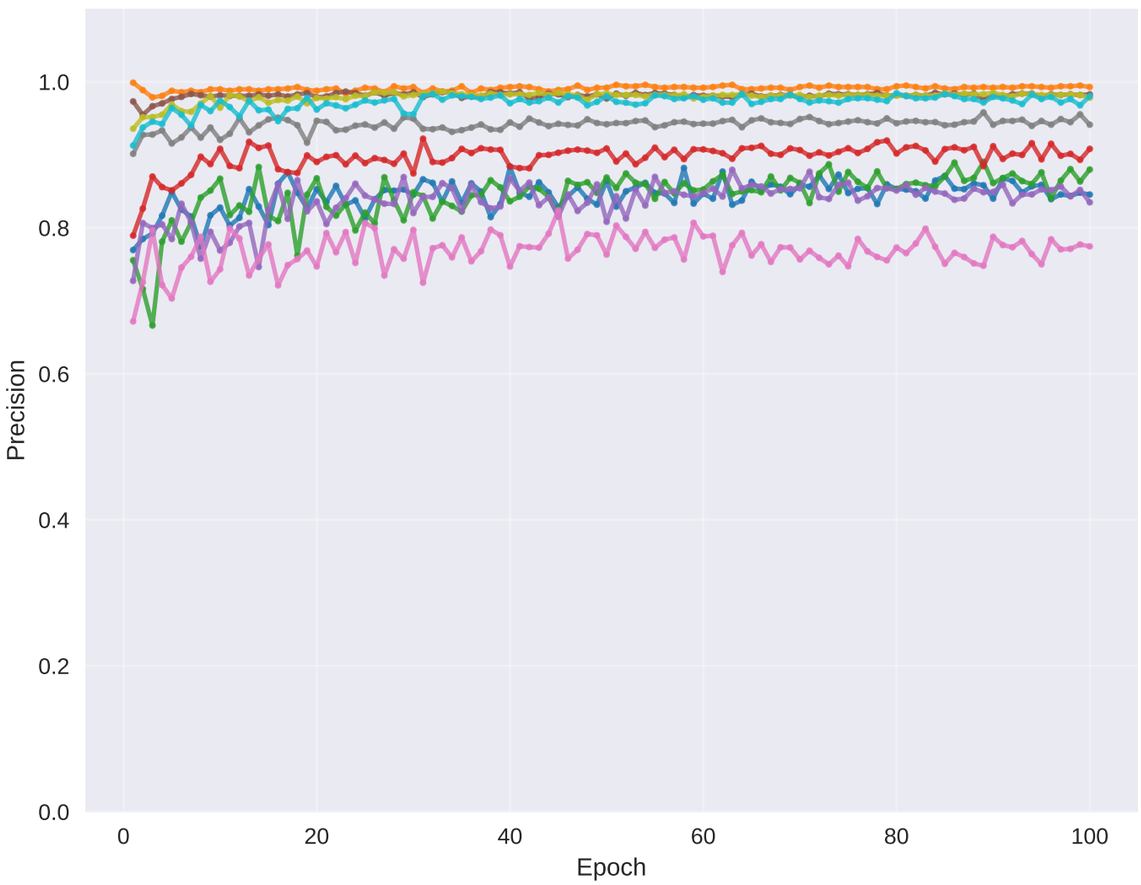



 Distribuzione Accuracy

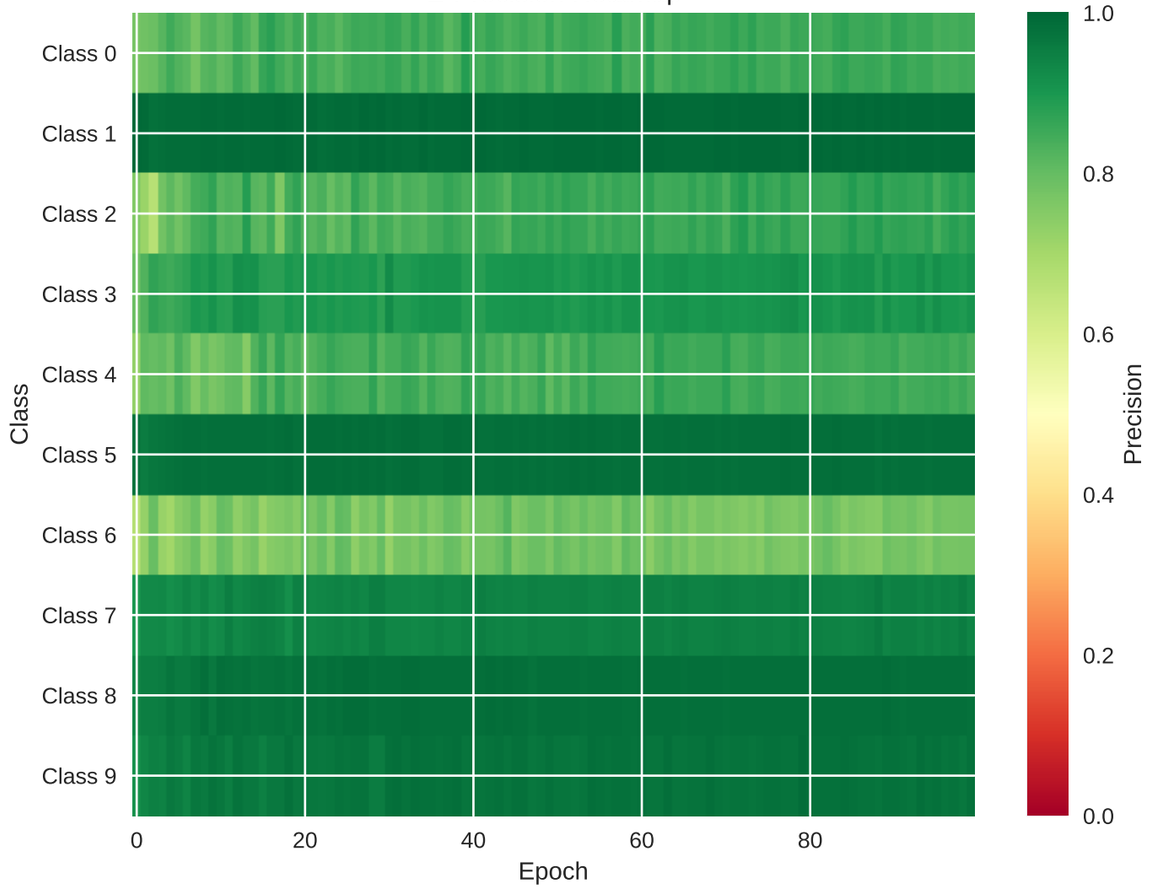


# Per-Class Precision Analysis

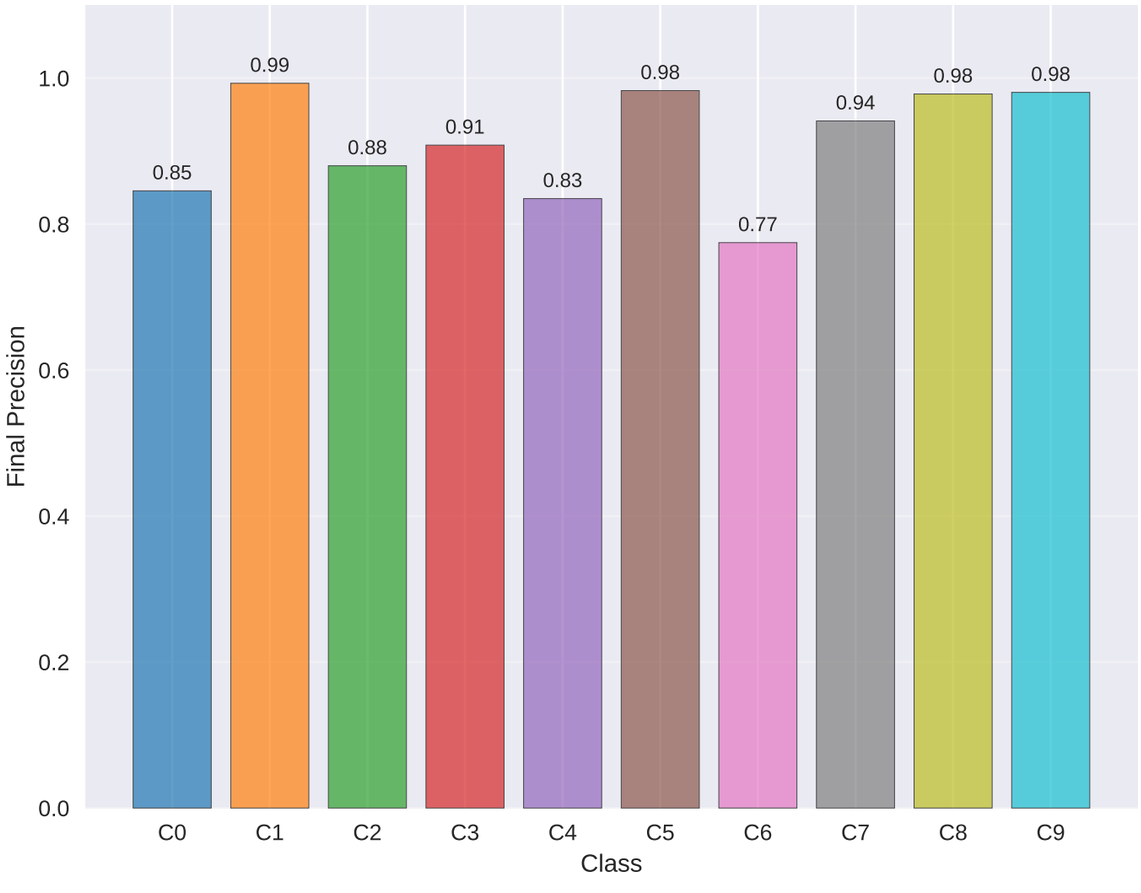
□ Per-Class Precision Evolution



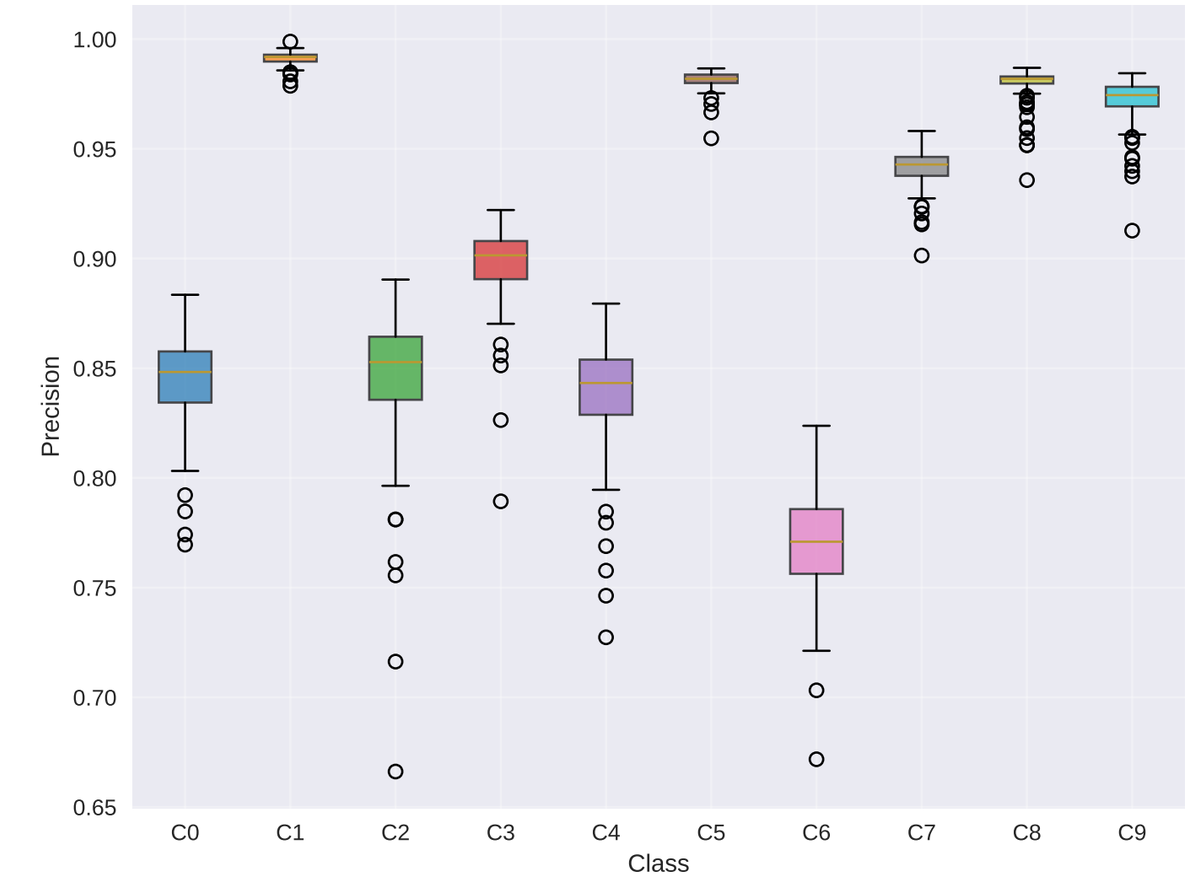

□ Precision Heatmap



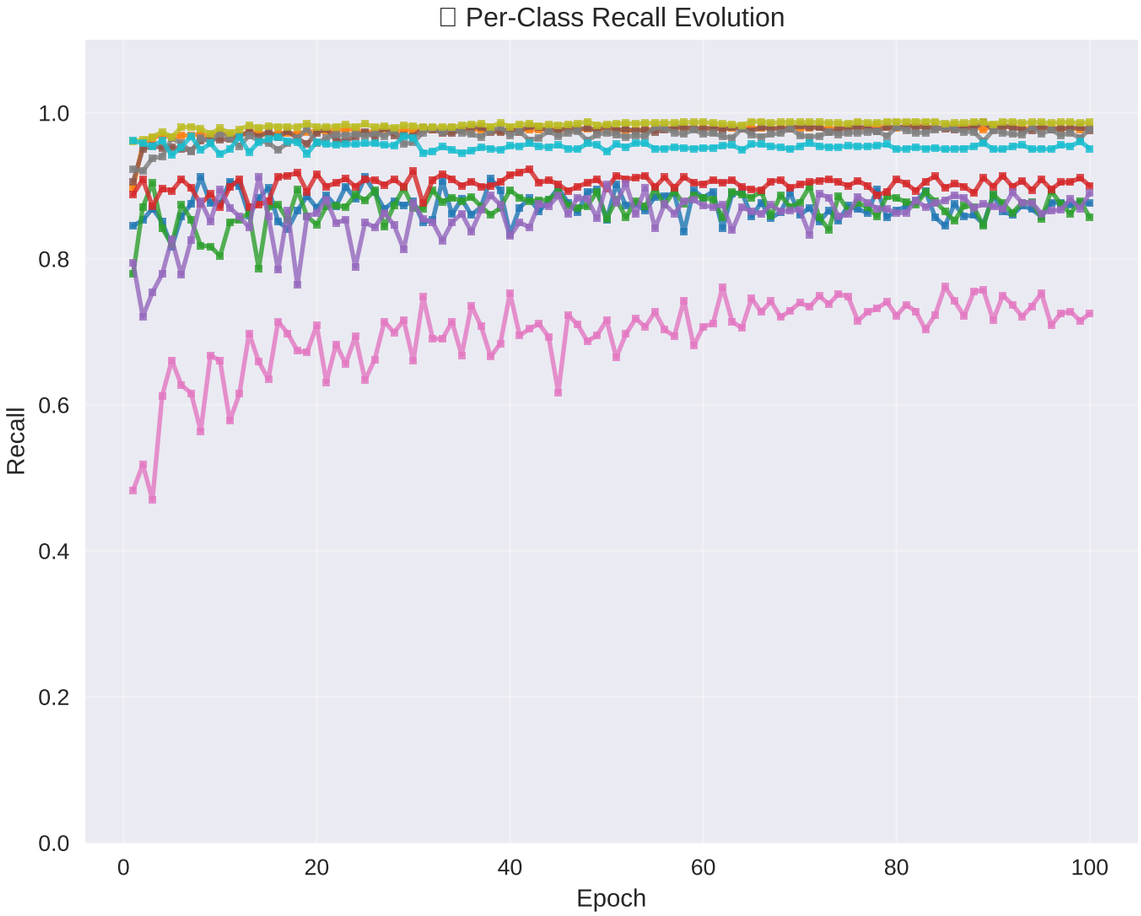

□ Final Precision per Class

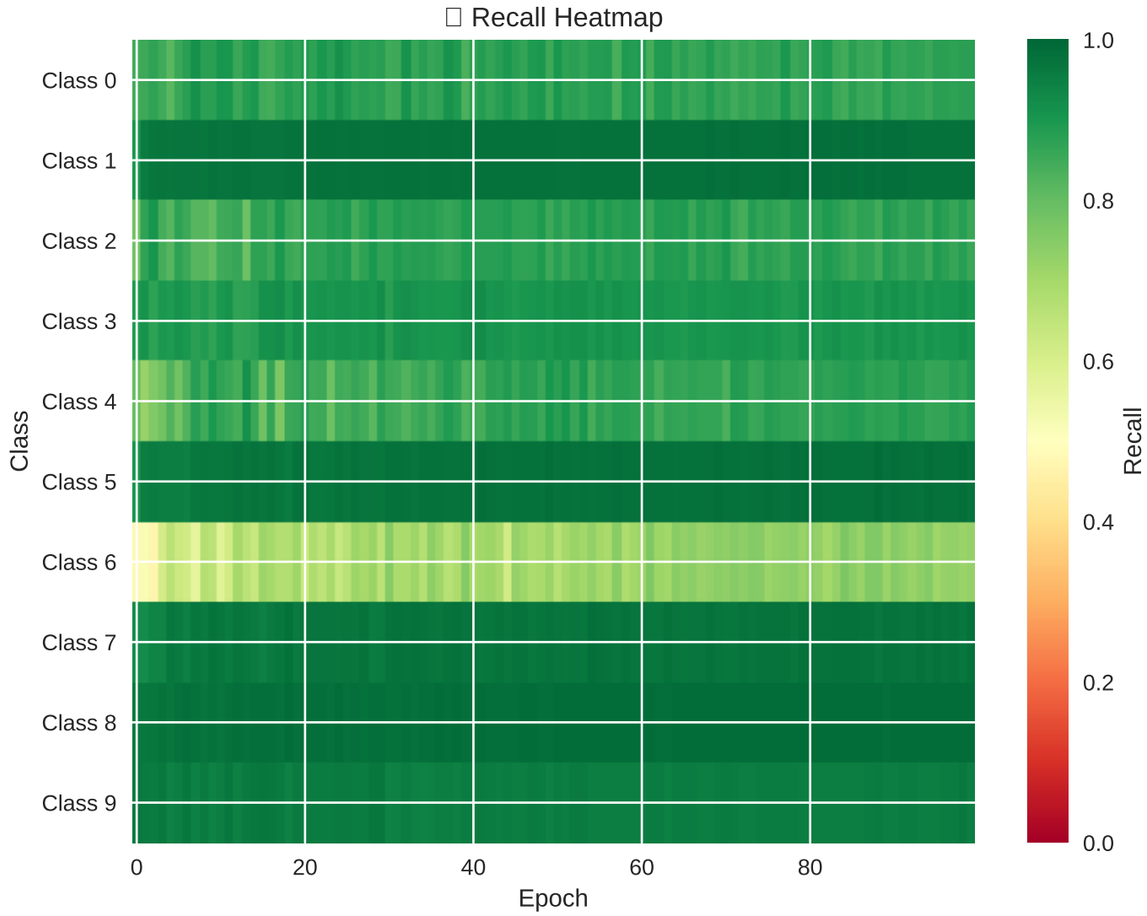


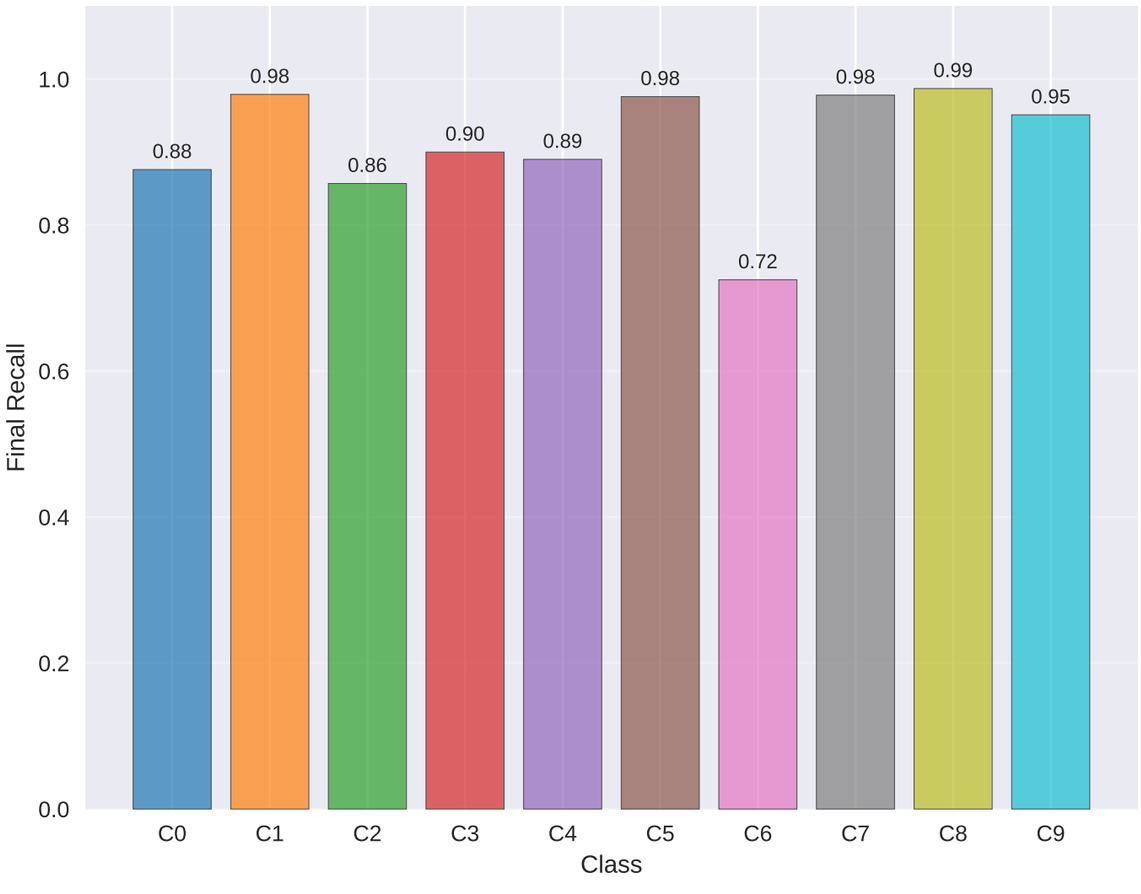

□ Precision Distribution per Class

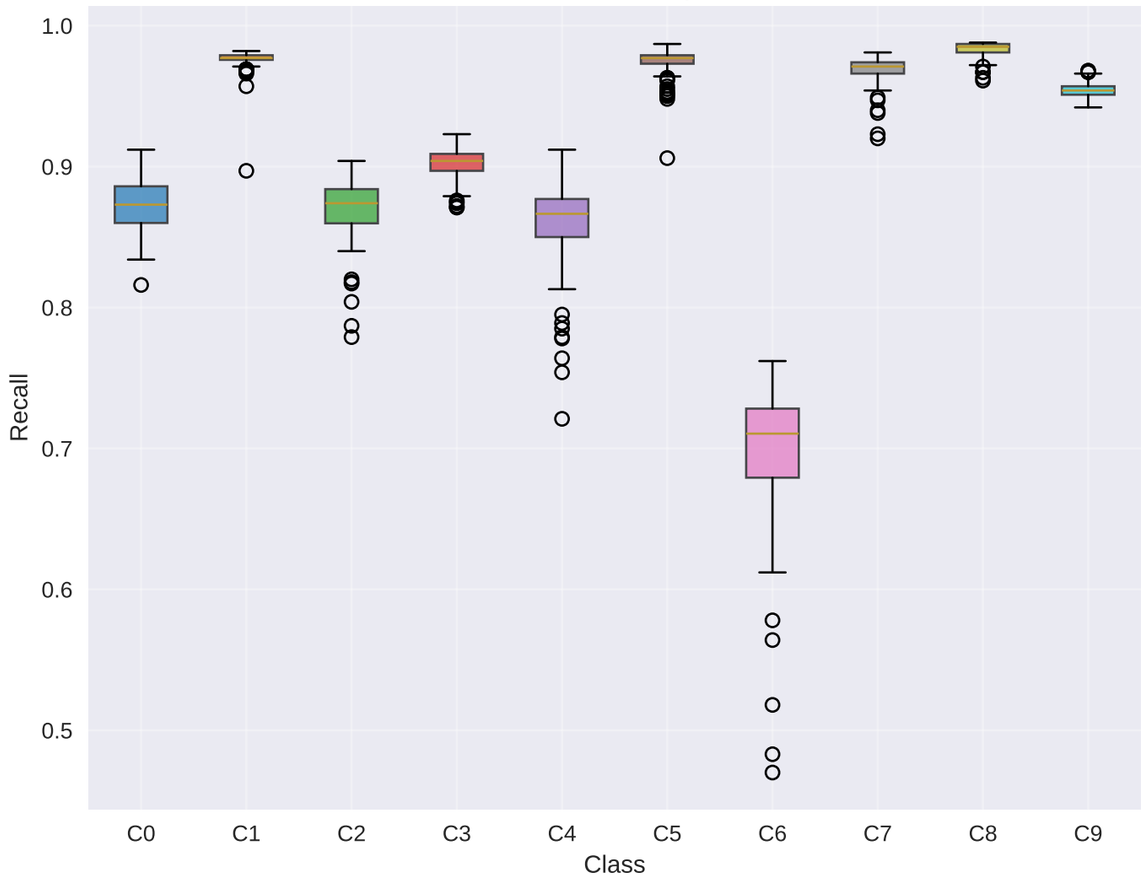


# Per-Class Recall Analysis

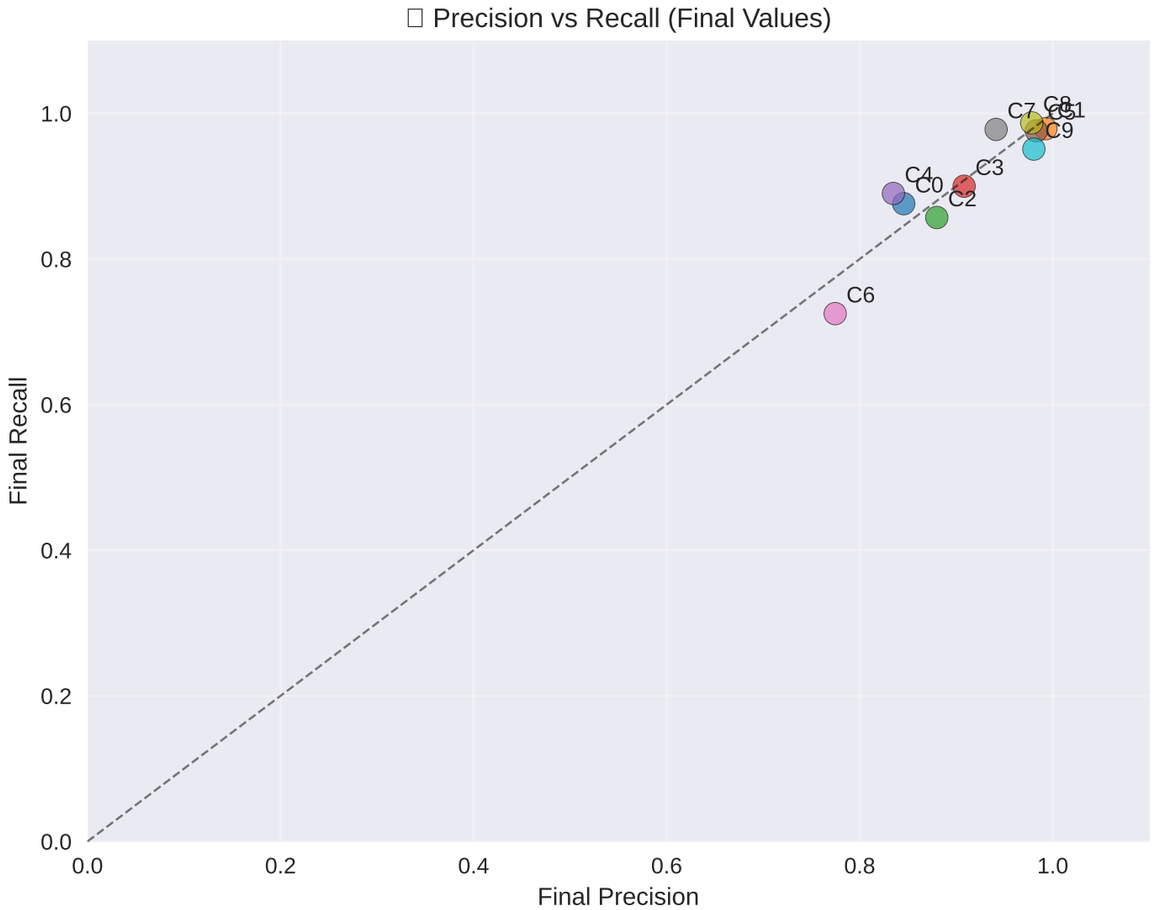


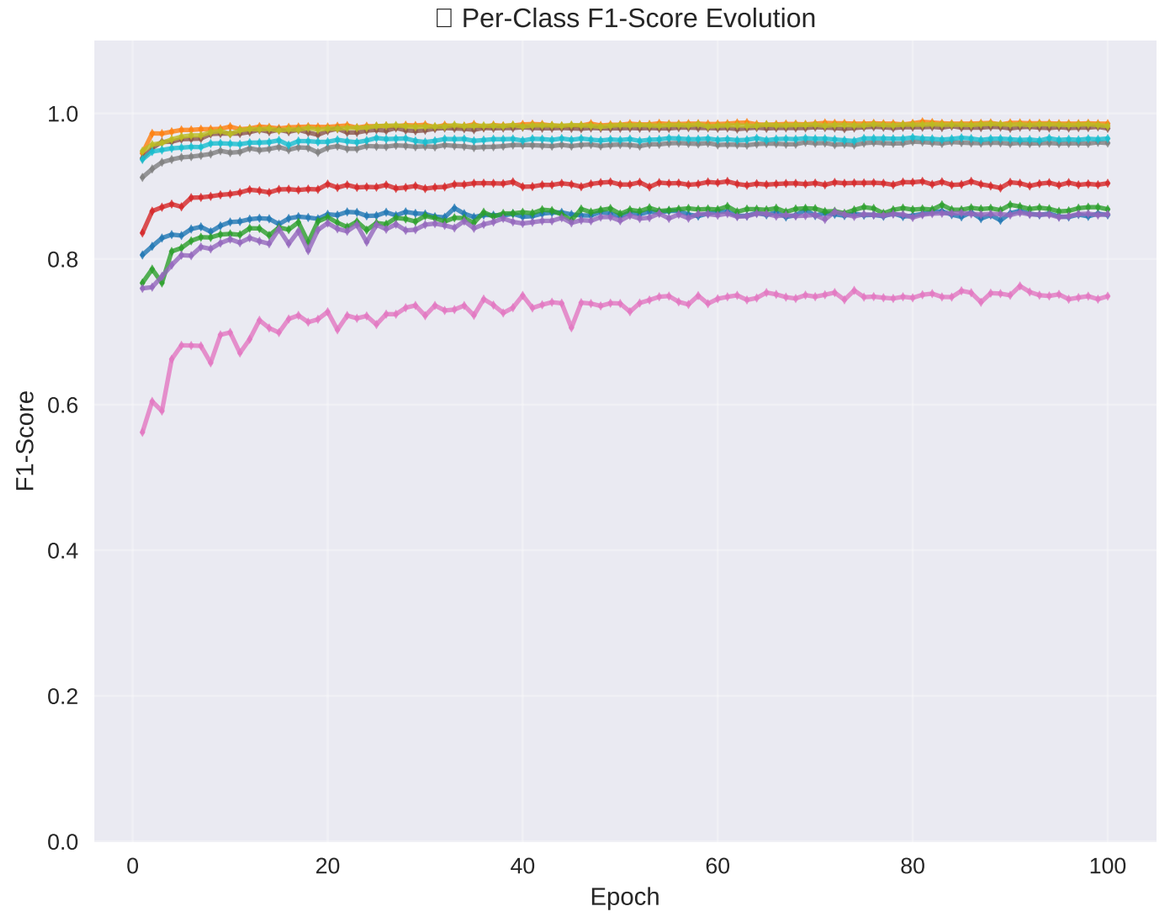



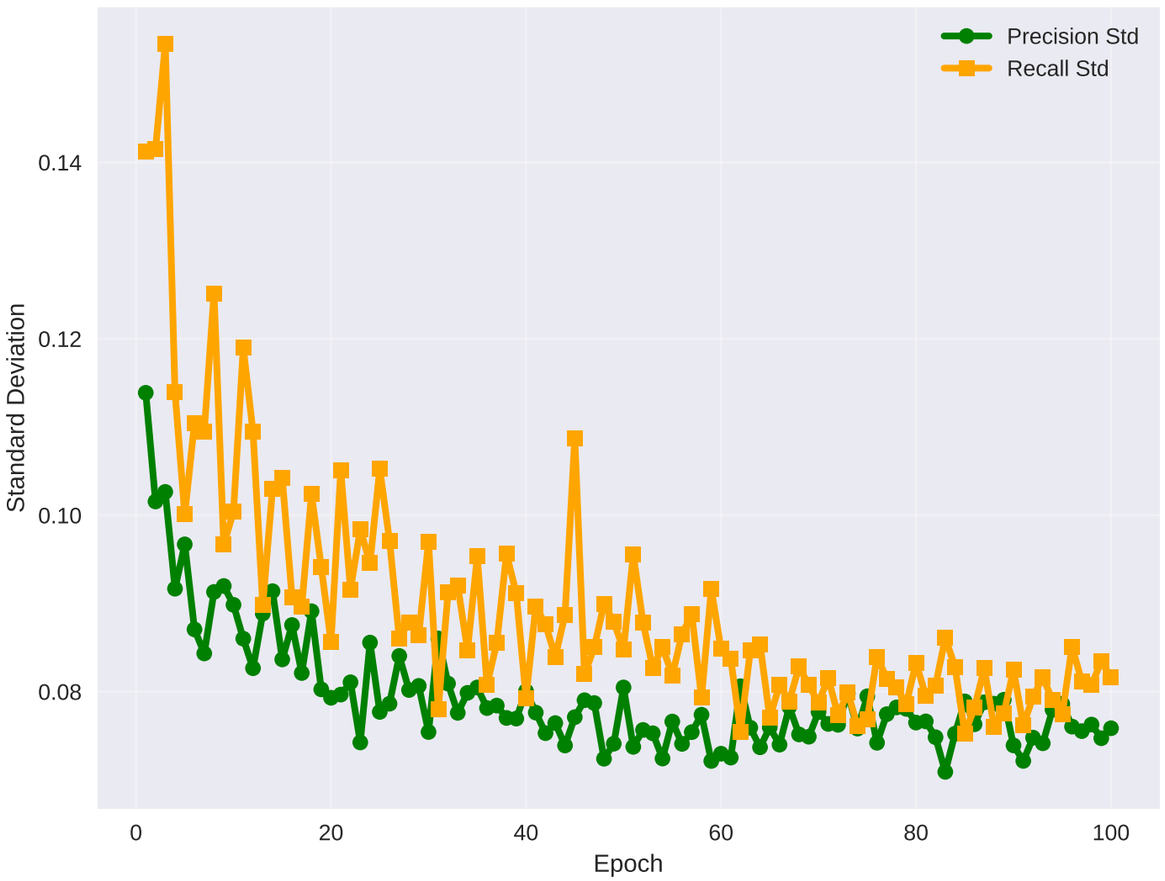



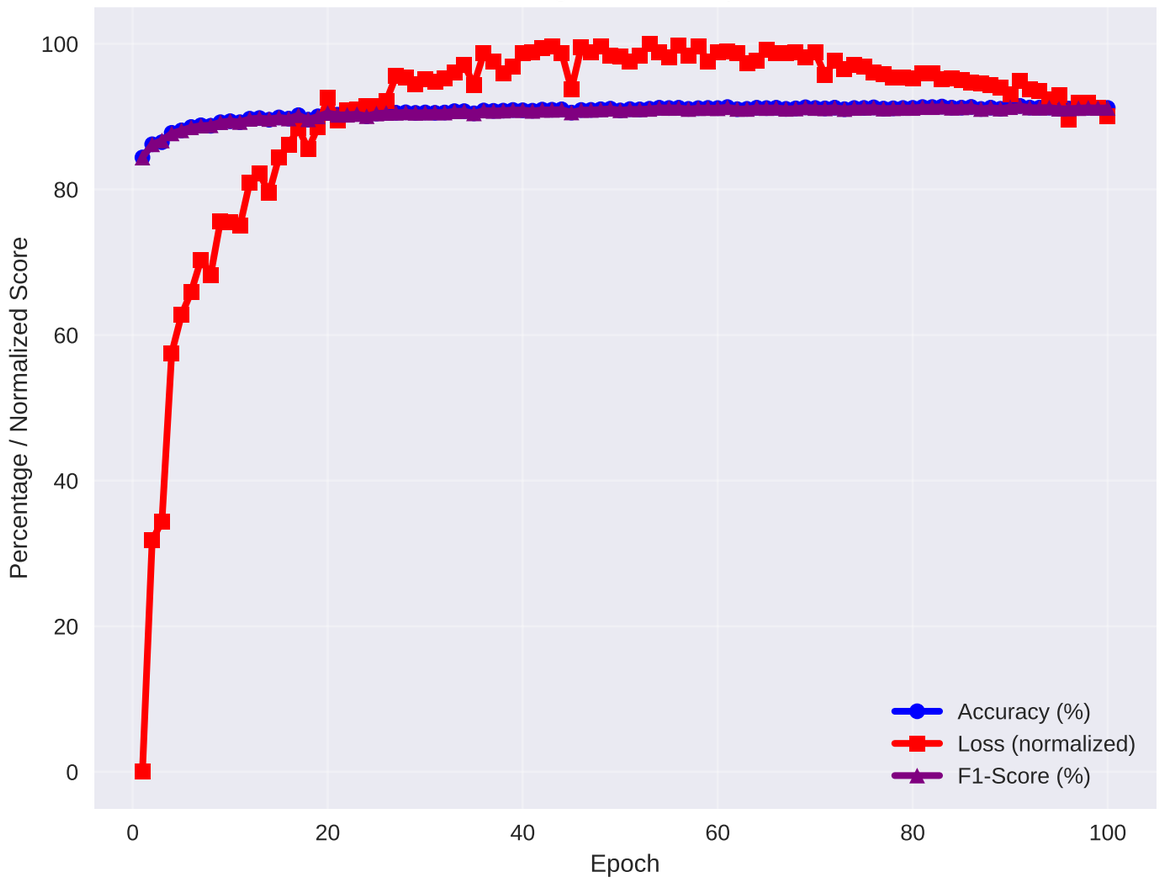



# Comparative Analysis



# Performance Summary Report

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FEDERATED LEARNING - PERFORMANCE SUMMARY
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FILE ANALIZZATO: 3000_results.csv
DATA ANALISI: 2025-09-05 15:59:43
EPOCHE TOTALI: 100

METRICHE FINALI:
• Accuracy: 91.19%
• Loss: 2.7325
• Average Precision: 0.912
• Average Recall: 0.912
• Average F1-Score: 0.912

EVOLUZIONE TRAINING:
• Miglioramento Accuracy: +6.80%
• Riduzione Loss: +1.7308
• Stabilità Accuracy (CV): 0.0126

PERFORMANCE PER CLASSE:
• Migliore Precision: Classe 1 (0.993)
• Peggiore Precision: Classe 6 (0.775)
• Migliore Recall: Classe 8 (0.987)
• Peggiore Recall: Classe 6 (0.725)

ANALISI DETTAGLIATA PER CLASSE:
• Classe 0: P=0.846, R=0.876, F1=0.861
• Classe 1: P=0.993, R=0.979, F1=0.986
• Classe 2: P=0.880, R=0.857, F1=0.868
• Classe 3: P=0.908, R=0.900, F1=0.904
• Classe 4: P=0.835, R=0.890, F1=0.862
• Classe 5: P=0.983, R=0.976, F1=0.979
• Classe 6: P=0.775, R=0.725, F1=0.749
• Classe 7: P=0.941, R=0.978, F1=0.959
• Classe 8: P=0.978, R=0.987, F1=0.983
• Classe 9: P=0.980, R=0.951, F1=0.965

VALUTAZIONE QUALITATIVA:
• Performance Globale: BUONA
• Stabilità Training: STABILE
• Balance Classes: SBILANCIATO

RACCOMANDAZIONI:
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